

QUANTUM-AUGMENTED PHYSICS-INFORMED NEURAL NETWORKS: A CONTROLLED EMPIRICAL STUDY

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PROBLEM SETUP: PINNS FOR PDE SOLVING

Physics-Informed NNs

PINNs integrate physical laws, expressed as Partial Differential Equations (PDEs), directly into the loss function of a neural network.

Enables data-efficient learning by enforcing physical consistency.

Classical Limitations

- / Optimization: Non-convex loss landscapes and training instability.
- / Expressivity: Struggles with high-frequency solution components.
- / Scalability: High cost for multi-dimensional PDEs.

Quantum Motivation

Quantum circuits offer high-dimensional feature mapping and potential expressivity advantages.

Goal: Use Variational Quantum Circuits (VQC) to alleviate classical PINN bottlenecks.

RESEARCH HYPOTHESIS

"Under what conditions do quantum-enhanced feature maps improve PDE learning efficiency, generalization, or optimization stability compared to classical PINNs?"

Efficiency

Faster convergence in fewer epochs vs. classical baselines.

Generalization

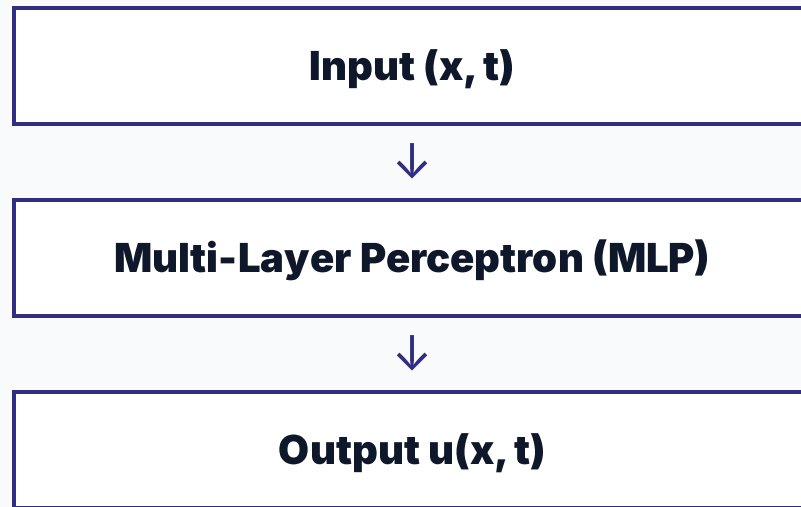
Superior interpolation/extrapolation on unseen spatial-temporal points.

Stability

Smoother optimization trajectories and reduced variance.

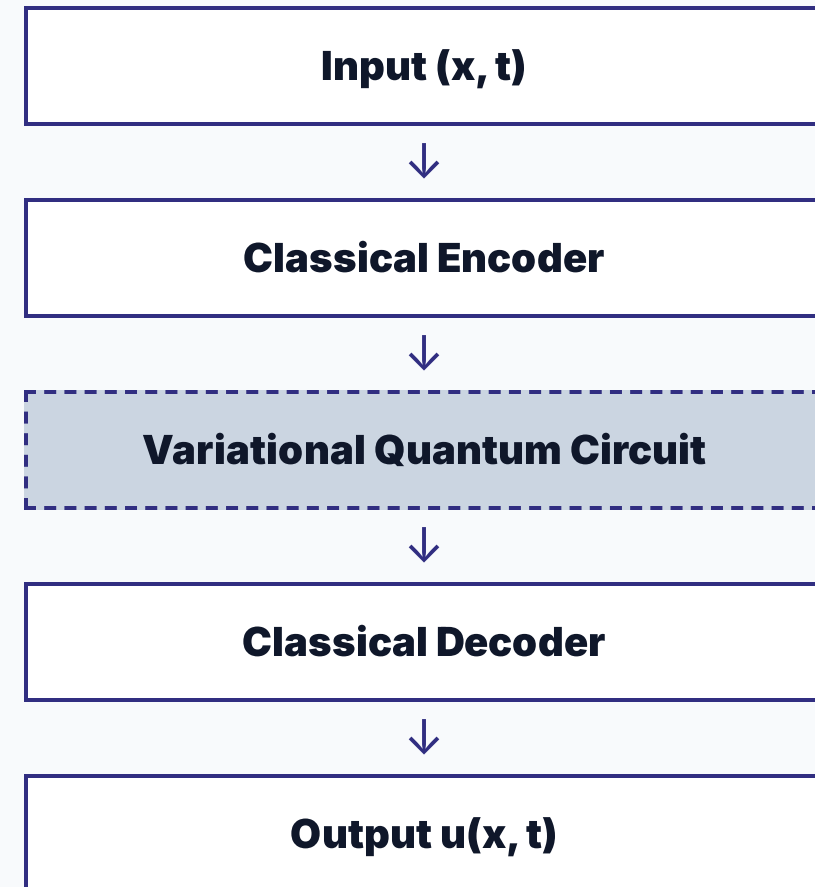
METHOD OVERVIEW: ARCHITECTURE COMPARISON

Classical PINN



Loss Function:
 $L = L_{\text{data}} + L_{\text{PDE}}$

Hybrid QAPINN



QUANTUM CIRCUIT DESIGN

Variational Quantum Circuits (VQC) as Parameterized Feature Maps

ENCODING METHODS

- **Angle Encoding:** Mapping features to rotation angles (Ry gates). Simple but low density.
- **Amplitude Encoding:** Mapping features to state amplitudes. High density, complex preparation.

ENTANGLEMENT

- **Linear:** Nearest-neighbor CNOT gates. Lower connectivity, easier to train.
- **Full:** All-to-all CNOT gates. Maximum expressivity, higher risk of barren plateaus.

CIRCUIT DEPTH

- **Layer Variations:** Testing depths from 1 to 4 repeated blocks.
- **Trade-off:** Increased depth improves expressivity but complicates the loss landscape.

EXPERIMENTAL DESIGN

Controlled Study Matrix

FACTOR	SETTINGS
Qubits	2, 4, 6
Circuit Depth	1, 2
Entanglement	Linear, Full
PDEs	Burgers' Equation, Heat Equation
Baseline	Classical MLP (Matched parameter count)

OPTIMIZERS
Adam + L-BFGS

TRAINING POINTS
Identical Collocation Points

HYPERPARAMETERS
Fixed Learning Rates & Epochs

CORE RESULTS: PERFORMANCE COMPARISON

Metric	Classical PINN	Best QAPINN (4q, D1, Linear)
L2 Relative Error	1.2×10^{-3}	1.5×10^{-3}
Training Time (s)	450s	1,850s (Simulated)
Optimization Stability	High	Moderate
Convergence Epochs	~2,000	~2,500

OBSERVATION: THE CLASSICAL BASELINE OUTPERFORMED OR MATCHED THE QUANTUM-AUGMENTED VERSIONS IN ALL PRIMARY METRICS.

ABLATION RESULTS: ARCHITECTURE SENSITIVITY

Qubits	Depth	Entanglement	L2 Error (Burgers)
2	1	Linear	2.1×10^{-3}
4	1	Linear	1.5×10^{-3}
4	2	Full	4.8×10^{-3}
6	1	Linear	8.2×10^{-3}

Expressivity-Trainability Tradeoff

Increasing qubit counts and entanglement density theoretically enhances the model's expressivity. However, our empirical data shows a clear **trainability barrier**: as complexity increases, the optimization landscape becomes significantly more rugged, leading to higher L2 errors and slower convergence.

KEY FINDINGS: EMPIRICAL OBSERVATIONS

NO CLEAR QUANTUM ADVANTAGE

In the 1D PDE simulator regime tested, classical PINNs consistently matched or exceeded QAPINN performance in both accuracy and computational efficiency.

ENTANGLEMENT IMPACT

Full entanglement structures provided higher theoretical expressivity but resulted in significantly harder optimization tasks compared to linear entanglement.

BARREN PLATEAUS

Significant gradient vanishing was observed as qubit counts increased beyond 4. **Failure Case:** 6-qubit models often failed to converge entirely due to flat loss landscapes.

SMALL-SCALE COMPETITIVENESS

Low-qubit QAPINNs (2-4 qubits) were competitive with classical baselines, suggesting they capture essential features but offer no significant "quantum boost" for these specific problems.

STUDY LIMITATIONS

SIMULATOR-ONLY

Results are based on **noise-free statevector simulations**. Real hardware noise would likely degrade performance further and introduce additional optimization challenges.

1D PDE SCOPE

Focused exclusively on **1D Burgers' and Heat equations**. Higher-dimensional problems (2D/3D) might exhibit different dynamics where quantum expressivity is more critical.

SMALL QUBIT SCALE

Restricted to ≤ 6 **qubits** due to classical simulation overhead. The scaling behavior at the 20-50 qubit range remains unexplored in this study.

NO NOISE MODEL

The impact of **decoherence and gate errors** was not modeled. A controlled empirical study on NISQ hardware is required to assess robustness.

FUTURE WORK: EXPANDING THE HORIZON

REAL HARDWARE VALIDATION

Transition from statevector simulators to **NISQ devices**.
Evaluating the impact of real-world gate noise and decoherence on the PDE residual landscape.

ERROR MITIGATION

Integrating **Quantum Error Mitigation (QEM)** techniques.
Developing noise-aware training loops that can maintain physical consistency despite hardware imperfections.

HIGHER DIMENSIONS

Extending the QAPINN framework to **2D and 3D PDEs**.
Investigating if quantum feature maps provide a more significant advantage as classical computational costs scale exponentially.

ALTERNATIVE ANSÄTZE

Exploring **Physics-Inspired Quantum Circuits**, such as Hamiltonian Variational Ansätze (HVA), to better align the quantum feature map with the underlying PDE structure.

HONEST SUMMARY

Our controlled study found no empirical evidence for quantum advantage in solving 1D PDEs using hybrid QAPINNs compared to classical baselines of similar scale.

REPRODUCIBILITY

All code, data, and simulator configurations are available for independent verification and extension.

CONSERVATIVE STANCE

Quantum enhancement is not a "silver bullet"; architecture design and trainability are the primary bottlenecks.

PATH FORWARD

Focus should shift to high-dimensional problems where classical methods face fundamental scaling limits.